The feature count as one inequality: a local payment and a global detection

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Abstract

The open cases of the feature-count theorem (relations, mixed slot-sets, compensation, $n \geq 3$) are not different problems. They are one inequality, split into two ingredients. The *local* ingredient bounds how much cheaper a single feature can be bought than its interference-free price, given its own sign-flipped coefficients. It involves one feature's neurons only, is finite-dimensional and does not depend on p. At $n = 2$ it is known sharply. The *global* ingredient says that the interference a network carries shows up in its worst input, unless the network pays for hiding it. All dependence on p, on the frequency set, on the other features and on compensation sits in this one statement. We state both, prove that together they give the count for every n, p, b, check that every case of the earlier notes is an instance, and reduce the global ingredient in the compensation-free class to a single statement about real trigonometric polynomials on $\mathbb{Z}_p^n$. References of the form "Note, X" are to `mechanism_note.tex`.

1 Setting

As in the Note: p an odd prime, $n \geq 2$ operands, $\nu = n+1$, $b \in [2, \nu]$, $s = \nu/b$, $r = 1/s$, $\rho = e^{2\pi i/p}$, $F = \{1, \dots, \frac{p-1}{2}\}$, $\kappa_l(\delta) = 1 - \cos(2\pi l\delta/p)$, $c_l(\delta) = 1 - \rho^{-l\delta}$ (so $|c_l(\delta)|^2 = 2\kappa_l(\delta)$). A network θ has neurons $f_i(x)[c] = W_i[c] \prod_t u_{it}(x_t)$; $P = \|\theta\|_{2,b}^b$; $g(x, \delta) = f(x)[y(x)] - f(x)[y(x) - \delta]$ with $y(x) = \sum_t x_t$; $h(\theta) = \min_{x, \delta \neq 0} g(x, \delta)$. For $l \in F$: clean coefficient β_l with atoms β_{il}, $Q_l = \sum_i |\beta_{il}|^r$, interference-free (IF) price $a_l = \nu^{1-s} Q_l^s$, cheap set $L' = \{l : |\beta_l| > a_l\}$. For $\emptyset \neq T \subseteq [n]$, $\beta_{l,T}$ is the sign-flipped coefficient of feature l's own atoms; it sits at the character $\chi_{l,T} = 2l\mathbf{1}_T \in \mathbb{Z}_p^n$ (Note, Theorem 11 and Lemma 6 of the paper: this holds for arbitrary inputs). $\Delta(\theta)$, $\mathcal{M}(\theta)$, $K_l(\mu)$, the unused budget $U(\theta)$ and the coverage gap $G(\theta)$ are as in the Note's section "Architecture of a proof of the count"; in particular $U + G = \nu^{1-s}(C_b P)^s \Gamma_s(p) - \min_\delta \sum_l a_l \kappa_l(\delta) \geq 0$.

2 The statement

Definition 1 (admissible payment)**.** A function $\Pi = \Pi_{n,s}$ of a vector $(\beta_T)_{\emptyset \neq T \subseteq [n]} \in \mathbb{C}^{2^n - 1}$ is an *admissible payment* if for every finite atom system (β_i, ξ_{it}) (one feature; $A = \sum_i \beta_i$, $Q = \sum_i |\beta_i|^r$, $\beta_T = \sum_i \beta_i \prod_{t \in T} \xi_{it}$)

$$\text{(L)} \qquad \operatorname{Re} A - \nu^{1-s} Q^s \;\leq\; \Pi\big((\beta_T)_T\big).$$

By the paper's Lemma 6 every feature of every network is such an atom system, so $\operatorname{Re} \beta_l - a_l \leq \Pi(\beta_{l,\cdot})$ for all l. (L) mentions one feature only; it does not depend on p, the width, or the other features.

Definition 2 (detection and the global ratio)**.** For a network θ with $L'(\theta) \neq \emptyset$ and $\mu \in \mathcal{M}(\theta)$ put

$$\mathrm{Det}_\mu(\theta) = \sum_\delta \mu_\delta \big(\mathbb{E}_x g(x,\delta) - \min_x g(x,\delta)\big), \qquad \mathrm{Pay}_\mu(\theta) = \sum_{l \in L'} K_l(\mu)\, \Pi(\beta_l, \cdot),$$

$$R_\Pi(\theta) = \sup_{\mu \in \mathcal{M}(\theta)} \frac{\mathrm{Det}_\mu(\theta) + U(\theta) + G(\theta)}{\mathrm{Pay}_\mu(\theta)}, \qquad \sigma^*_\Pi(n,p,b) = \inf\{R_\Pi(\theta) : L'(\theta) \neq \emptyset\}.$$

The *global ingredient* is

$$\text{(G)} \qquad R_\Pi(\theta) > 1 \quad \text{for every network } \theta \text{ with a cheap feature.}$$

Theorem 3 (the two ingredients give the count)**.** *Fix n, p, b and an admissible payment Π. If (G) holds, then every global maximiser of the $L_{2,b}$-margin uses exactly $|\operatorname{spt} y^\star|$ features for some maximiser $y^\star$ of $\Gamma_s(p)$. In particular this holds if $\sigma^*_\Pi(n,p,b) > 1$.*

Proof. Let θ have a cheap feature. By (G) pick μ with $\mathrm{Det}_\mu + U + G > \mathrm{Pay}_\mu$. By (L), $\sum_l (\operatorname{Re}\beta_l - a_l) K_l(\mu) \leq \mathrm{Pay}_\mu$ (non-cheap features contribute ≤ 0, and $K_l \geq 0$). Taking point masses at the minimisers of $g(\cdot,\delta)$ as tests in the master inequality (Note, Proposition "master inequality") gives the strict form of (MI), so $\gamma(\theta) < \nu^{1-s} B_p \Gamma_s(p) \leq \gamma^*_b$. Hence no global maximiser has a cheap feature, and Note, Proposition "no cheap features $\Rightarrow$ exact count" finishes. □

Remark 4 (what each side carries)**.** Pay is linear in the features: each cheap feature contributes its own payment, weighted by how much it covers μ. Det is the gap between mean and worst input of the actual margin profile. It sees every mode of g: the cheap features' own interference, the other features' interference, partner and foreign modes (compensation), and all coincidences between them. U pays for norm that buys no clean amplitude (in particular for paid compensation, Note, Lemma "paid compensation is quadratic"), and G pays for an inefficient allocation. So (G) is a single statement about how interference, compensation and coverage trade off. The global ratio is measured in units of the local payment: $\sigma^*_\Pi > 1$ means detection beats the best conceivable discount.

Remark 5 (the price need not be the IF price)**.** Note, Proposition "decoupling suffices" shows that the count needs only decoupling with a *common* price. Replacing $\nu^{1-s}Q^s$ in (L) and in a_l, U, G by $\pi_{n,b}(p)Q^s$, the best single-feature price, gives a version (L_π)+(G_π) of Theorem 3 with the same proof. Single-feature cheats are then absorbed into the price, which can only lower Π. This is the form to use for $n \geq 3$, where the IF price may not be the true single-feature price (open problems O3, O4 of the Note).

3 Coverage: every case is an instance of (G)

(G) quantifies over all networks, so nothing is left out. The table records how the earlier cases sit inside it, and which parts of (G) are proved. "Line" means that all cheap features' interference sits on characters of one slot-set T, so that it depends on x through one linear form $s_T(x)$ only; "plane" (and higher) means several slot-sets.

class of networks	what (G) becomes	proved
interference-free	vacuous ($L' = \emptyset$)	all n, p, b (Note, no-cheap-features proposition)
one frequency, any inputs	single cheap feature, cone geometry	$n = 2$: every p (Note, Thm. "price of a feature"); every n: large p
class $\mathcal{C}$, line ($T = \{1,2\}$, $n = 2$)	trig polynomial on $\mathbb{Z}_p$ with spectrum $2S$ (Prop. 6)	$p \le 59$; any p if $\sum_{L'} \kappa < 29.7$; near-IF; inefficient allocations
class $\mathcal{C}$, plane ($n = 2$, several T)	trig polynomial on $\mathbb{Z}_p^2$	$p = 5, 7$; few cheap features; (D_3)-dissociated cheap sets; near-IF
compensation (outside $\mathcal{C}$)	extra modes at $\chi_{l,T}$ inside g; paid part in U	single cheap feature: $p = 5, 7, 13$ proved, $p \le 41$ numerical
$n \ge 3$	same, with Π_n	Π_n finite (crude); dissociated sets if $k^\flat(n, s) < 1$
$p = 3$	(G) *fails*	the counterexamples; see below

Consistency at $p = 3$. The certified $p = 3$ networks beat the IF value, so (G) must fail there, and it does in the place the statement predicts: their interference is hidden by compensation through the coincidence $2 \equiv -1 \pmod 3$ (constant input components put a mode exactly on $-\chi_{1,T}$; Note, remark after the Riesz-product theorem). The local ingredient (L) holds at $p = 3$ (it does not depend on p). So the only way the count could fail at $p \ge 5$ is the same way: a global failure of detection.

4 The compensation-free class: one statement about trigonometric polynomials

Proposition 6 (class $\mathcal{C}$)**.** *Let $\theta \in \mathcal{C}$, a_l its IF prices, $e(a) = \min_\delta \sum_l a_l \kappa_l(\delta)/(\Gamma_s \|a\|_{1/s})$ its efficiency, $\delta_0 \in \Delta(\theta)$, and $\mu = \frac{1}{2}(\mathbf{1}_{\delta_0} + \mathbf{1}_{-\delta_0})$. Then*

$$\mathrm{Det}_\mu = \tfrac{1}{2}\big(-\min_x F_{\delta_0}(x) - \min_x F_{-\delta_0}(x)\big), \qquad F_\delta(x) = \sum_{l \in F} \sum_{T \ne \emptyset} |\beta_{l,T}|\, |c_l(\delta)| \cos\Big(\tfrac{2\pi}{p}\, 2l\, s_T(x) + \psi_{l,T} + \arg c_l(\delta)\Big),$$

with phases $\psi_{l,T}$, and $G(\theta) \ge (1/e(a) - 1) \sum_l a_l \kappa_l(\delta_0)$, $|\beta_{l,T}| \le \nu^{s-1} a_l$.

Proof. The proof of Note, Lemma "interference profiles are slack potentials" gives $g(x, \delta) = \mathbb{E}_x g(\cdot, \delta) + F_\delta(x)$ with $F_\delta(x) = \sum_{l,T} \mathrm{Re}\,\big(a_{l,T} c_l(\delta) \rho^{2l s_T(x)}\big)$ and $|a_{l,T}| = |\beta_{l,T}|$. The identity for G is its definition with $\|a\|_{1/s} = \nu^{1-s}(\sum Q_l)^s$, after dropping $U \ge 0$; $|\beta_{l,T}| \le \sum_i |\beta_{il}| \le Q_l^s$. □

So, in class $\mathcal{C}$, (G) follows from the following statement, in which no network appears.

Conjecture 7 (no unicorns, class $\mathcal{C}$)**.** *For every allocation $a \ge 0$ on F, every δ_0 minimising $\sum_l a_l \kappa_l$, every set $L' \subseteq \operatorname{spt} a$ and every family $(\beta_{l,T}, \psi_{l,T})$ with $\beta_{l,T} = 0$ for $l \notin L'$, $0 < \max_{l,T} |\beta_{l,T}|$ and $|\beta_{l,T}| \le \nu^{s-1} a_l$,*

$$\tfrac{1}{2}\big(-\min F_{\delta_0} - \min F_{-\delta_0}\big) + \big(\tfrac{1}{e(a)} - 1\big) \sum_l a_l \kappa_l(\delta_0) \;>\; \sum_{l \in L'} \kappa_l(\delta_0)\, \Pi(\beta_{l,\cdot}).$$

At $n = 2$ the admissible payments are $\Pi = c_1 \max_T |\beta_T|$ with $c_1 = 1 - 3^{1-s}$ (sharp, Note, Thm. "sharp pointwise payment"), and $\Pi = c_{12} |\beta_{\{1,2\}}|$ with $c_{12} = 1 - (\frac{2}{3})^{s-1}$ when only $T = \{1, 2\}$ is present (Note, P_{12}). The conjecture is homogeneous except for the efficiency term. At an optimal allocation ($e = 1$) it says: the worst input sees more interference than the best discount. Two facts already in the Note show where it is delicate:

- *Typical versus worst input.* For $|S| = K$ frequencies the root-mean-square of F_δ is of order $\sqrt{K}$ times a coefficient, while the payment is of order K times a coefficient. The conjecture therefore needs the worst of the p^n inputs to see order K. With $K \approx 1.4 \ln p$ there are $p \approx e^{0.7K}$ inputs per line, enough for large deviations of order K when the frequencies are spread out (measured: worst input $\geq 0.54K$ with adversarial phases at $p = 10007$, $K = 13$, against $0.18K$ needed in the line geometry).

- *Structure.* A set with many short additive relations (a progression) admits flat polynomials (measured $0.17K$), but covers badly ($e \approx 0$). A unicorn would be a set that is both an efficient cover and flat.

5 Measurement protocol

For programme supports $S = \operatorname{spt} y^\star$ at $n = 2$, $b = 2$ (so $e = 1$), binding offsets δ_0, and each geometry, compute

$$\rho_{\mathrm{geom}}(p, S, \delta_0) = \min_{(\beta_{l,T}, \psi_{l,T})} \frac{\frac{1}{2}(-\min F_{\delta_0} - \min F_{-\delta_0})}{\sum_l \kappa_l(\delta_0)\Pi(\beta_{l,\cdot})}$$

by adversarial search (smoothed minimum, random and structured restarts). Geometries: line $T = \{1, 2\}$ with $\Pi = c_{12}|\beta_{12}|$; line $T = \{1\}$ with $\Pi = c_{12}|\beta_1|$ (proved below, Section 6); plane (all T) with $\Pi = c_1 \max_T |\beta_T|$. Fixing δ_0 only helps the network, so each measured value is a lower bound for the corresponding restriction of σ^*_Π. The conjecture at $e = 1$ needs $\rho > 1$.

6 First push: configurations, corrections and the refined crux

Correction: the flatness that matters is two-sided. In the line geometry the detection at δ_0 is $\frac{1}{2} \operatorname{osc}_w D_{\delta_0}(w)$ (Note, Lemma "oscillation"), half the peak-to-trough range of a cosine sum with free phases. Its Sidon-type constant $\sigma_2(S) = \inf_{u \geq 0, \theta} \frac{1}{2} \operatorname{osc}_w \sum_{l \in S} u_l \cos(2\pi l w/p + \theta_l) / \sum u_l$ obeys Parseval, $\sigma_2 \geq 1/\sqrt{2K_{\mathrm{eff}}}$. The one-sided analogue (depth of the deepest dip) is the wrong object: glued Fejér-weighted progressions are one-sided flat with good coverage (agent A, `agents/A_result.md`).

Measured global ratios (Section 5 protocol, adversarial, $b = 2$; `mechanism_scripts/unified/`). Line geometry with the sharp constant: 3.9–5.1 on programme supports for $101 \leq p \leq 2003$ (most pessimistic constant c_1: 1.7–2.2). Plane geometry with $\Pi = \min(c_1 \max_T, \|\cdot\|_2/2\sqrt{3})$: 2.6–3.2 for $p \leq 151$. The cheater needs < 1.

Building the cheater (`unified/unicorn/`). (i) *The detection side turns at small size:* a block of 32 frequencies placed antipodally to a binding offset ($l\delta_0 \equiv \frac{p-1}{2} - j$) and phased to be flat has half peak-to-trough $0.175 \cdot \sum u < c_{12}$. (ii) *Embedded in the programme support at $p = 10007$ it loses at every budget share, even with interference free:* the value is $(1-f)^{3/2}$ times the optimum, i.e. the block adds nothing to the worst case. The bottleneck moves to a tied-worst wrong answer where the block is nearly blind (strength 0.18 against average 1). (iii) Re-optimising all weights, placing the block to be strong (≥ 0.89) at all 480 worst wrong answers of a 120-frequency cover ($p = 30011$), or adding 100 clean "patching" frequencies does not help: the optimiser drops the block ($\times 0.950$ to $\times 0.985$), while a random block of the same size gains $\times 1.088$ with free interference. The flat block's own help profile has minimum 0.000 (structural blind spots); the random block's is 0.56.

The trilemma. A profitable dirty block must (a) cover on its own, since once it takes budget its blind spots become the bottleneck; (b) be two-sided flat; (c) be small in the r-participation sense, since thin layers are paid for by the concave cost. Random sets have (a),(c); progressions (b),(c); glued flat blocks (a),(b) but need many frequencies: with optimised Rudin–Shapiro-type blocks, $\sigma_2 \le 0.18$ with coverage t needs $K \approx 11.6 \ln p$ ($t = 0.3$) to $22.4 \ln p$ ($t = 0.6$) as $p \to \infty$, against $K^* \approx 1.4 \ln p$ (agent A, `agents/A_result2.md`; best explicit witness $p = 100003$, $t = 0.36$, $K = 8.3 \ln p$, $\sigma_2 = 0.173$). Inside the window $K \le 2.5 \ln p$ with coverage ≥ 0.3, the smallest σ_2 found is 0.36, and programme supports with free weights give 0.357–0.551. The conjecture is that no set has all three.

Proved in this push (agents B, C; proofs in `agents/`).

- *Slot symmetry* (checked here). With $\beta_i = r_i e^{i\alpha_i}$, $\xi_i = e^{i\theta_i}$, $\eta_i = e^{i\phi_i}$ and $(\omega_0, \omega_1, \omega_2) = (\alpha + \frac{\theta+\phi}{2}, -\frac{\theta}{2}, -\frac{\phi}{2})$, the coefficients $A, B, C, \bar{D}$ are the sign flips of $\omega.$ in none, slot 1, slot 2, slot 0. Permuting the ω's atomwise is a symmetry that fixes $|A|, Q$ and permutes $(|D|, |B|, |C|)$. Hence P_{12} gives the sharp constant c_{12} for *every* single slot-set, and the ℓ^1 payment $c_{12}(|\beta_1| + |\beta_2| + |\beta_{12}|)$ is numerically sharp (proved on the axes and away from them; open: one dominant slot-set plus small others).

- *Only short relations matter* (B). A degree-D polynomial certificate sees only relations of length $\le D$; if S has no relation of length $\le 2n + 1$ then $\sigma_2(S)$ is at least the smallest Gauss node of the independent model, with no coverage hypothesis.

- *Size must be counted with order* < 1 (B). With $K_{\mathrm{eff}} = (\sum u)^2 / \sum u^2$ the clean lemma is false even two-sided (a Gauss-sum "thin layer" over all of F gives coverage with no oscillation). In the application the right size is the r-participation number, which the concave cost bounds.

- *Reductions* (C): thin blocks are closed; a block weak at a near-tied offset of the rest is closed for small shares (the lopsided-help mechanism); the plane reduces to the line with the same payment constant and at most $3\times$ the requirement on σ; compensation reduces to the line lemma plus a per-feature quadratic game whose constants depend on p.

The refined crux: stability. Everything now hangs on one statement about the programme:

Conjecture 8 (stability)**.** *If an allocation has efficiency $e(a) \ge e_0$ (about 0.76), the a-weighted mass of short additive relations among its support, of each length $D \le D_0$, is a small multiple of the independent baseline, uniformly over sub-blocks carrying interference.*

With it, "only short relations matter" gives $\sigma_2 \approx 0.31$–0.43 for $K / \ln p = 1.5$–2.5 (limit formula numerical), above $c_{12} = 0.18$, and closes the relations case at every prime. Partial certificates (agent A): moment certificates from exact relation sums of length ≤ 8, valid for all phases, give 0.27–0.34 on programme supports for $p \le 2003$ after local minimisation over the weights; under "relation mass below the independent baseline" the tilted mean gives $\sigma_2 \ge 0.28, 0.22, 0.18$ at $K / \ln p = 2, 2.5, 3$ asymptotically. The remaining sub-claim is a weighted Sidon (Pisier–Bourgain-type) inequality on near-optimal supports. Evidence: programme supports carry 2 three-term relations against 77.8 expected for random sets of the same sizes (83 of 85 supports have none), and 235 four-term relations against 468. Optimal covers repel short relations more strongly than chance.

7 Proof plan

1. *Clean lemma (line).* If the frequencies carrying interference cover $\mathbb{Z}_p$ with their own weights ($\min_{\delta\neq 0}\sum_l v_l\kappa_l(\delta) \geq t\sum_l v_l$) and their effective number is $O(\ln p)$, then every real polynomial with those frequencies and amplitudes has $-\min \geq \sigma(t)\sum_l v_l$ for an explicit $\sigma(t) > 0$ independent of p: a deterministic Sudakov minoration for short trigonometric polynomials on $\mathbb{Z}_p$.

2. *Blind spots.* Reduce arbitrary interference supports to the clean lemma, using the efficiency term: a sub-block that covers badly is blind on a large set of offsets, which the rest of the support must over-cover at a cost in $e(a)$.

3. *Plane to line.* $-\min$ over $\mathbb{Z}_p^2$ is at least $-\min$ over any line $\{(t, mt)\}$, on which the three slot-sets become the frequency sets $2S$, $2mS$, $2(1+m)S$. Average or choose m.

4. *Compensation.* Compensating modes enter F as extra coefficients at the same characters. Paid ones are charged to U; free ones need the coincidences (F1), (F2) of the Note, which are short relations of the same kind as in step 2.